

Transcendence Theory

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This is kinda just me trying to comprehend Schneider-Lang.

§1 Classical Transcendence Theory

A typical proof of irrationality/transcendence involves coming up with a quantity X that is:

1. nonzero
2. small (for analytic reasons)
3. discretized away from 0 (if our number were not irrational/transcendental)

To be a bit more precise, a specific quantity X will give us a lower bound on the height of our number, and we can establish irrationality/transcendence with a sequence of quantities giving an arbitrarily high lower bound on height.

Example 1. Suppose that e^3 was a rational number with denominator q . We show that q must be at least 286. Consider the following quantity:

$$X = \int_0^3 \frac{x^{10}(3-x)^{10}}{10!} e^{3-x} dx = 81(1741039699e^3 - 34969717159).$$

Our quantity X

1. is positive
2. is approximately 0.0034963
3. is a multiple of $1/q$.

Hence, q must be at least $1/0.0034963 > 286$.

Replacing 10 with an arbitrary large positive integer establishes the irrationality of e . We will extend this idea to prove that e is transcendental.

Theorem 2. For any integers $c_1, \dots, c_m$, not all zero, we have

$$c_1 e + \dots + c_m e^m \notin \mathbb{Z}.$$

In particular, e is transcendental.

Proof. Suppose, for the sake of contradiction, we had $c_1e + \dots + c_ke^k \in \mathbb{Z}$. Let $P(z) = z(z-1)\dots(z-k)$. Consider the quantity

$$X = \sum_{j=1}^k c_j \left(\int_0^j \frac{P(z)^N Q(z)}{N!} e^{j-z} dz \right),$$

where we allow $Q \in \{1, x, x^2, \dots, x^k\}$. Our quantity X

1. is nonzero infinitely often by, say, [Stone-Weierstrass](#)
2. is at most $C^N/N!$ for some constant C
3. is an integer (integration by parts, will be written out later).

Taking N sufficiently large, we obtain a contradiction. □

§2 Between Classical Transcendence and Schneider-Lang

Here's an important question: **what property of e^z did we use** to show that it outputs $\mathbb{Z}$ -independent numbers when z is an integer? What is the “reason” e is transcendental? According to Schneider-Lang, the reason is that, as a function, $z \rightarrow e^z$ is

1. algebraically independent of z
2. holomorphic and not too large
3. a solution to a differential equation $((e^z)' = e^z)$.

It's sort of obvious that e^n would need to be algebraically independent from n to be irrational. We don't actually really use the second point in the classical proof, and indeed, the classical proof doesn't obviously generalize to Schneider-Lang. Maybe it's just a happy coincidence that we can get a proof to work for e^z without too much work? Anyways, I'll try to justify that we used e^z being a solution to a differential equation in the classical proof.

Both the classical proof above and Schneider-Lang analyze fundamentally similar quantities. If we trace through the case of Schneider-Lang that proves that e can't be algebraic of degree k , we find that we are considering a function of the form

$$X(z) = \sum_{i,j} c_{i,j} z^i e^{jz},$$

where $i \ll N^{1-\varepsilon}$ and $j \ll N^\varepsilon$, and X is chosen to vanish up to degree N at $0, 1, \dots, k$. Our contradiction is obtained by

1. picking the smallest N' such that $X^{(N')}(0)$ is nonzero, which exists by the algebraic independence of e^z and z .
2. using complex analysis, leveraging the disappearing at $k+1$ points to prove $\left| X^{(N')}(0) \right| \ll (N!)^{1-k}$
3. using the fact that the derivatives of X are all algebraic to prove that $\left| X^{(N')}(0) \right| \gg (N!)^{1-d}$.

Compare this to the classical proof, where we do something similar except with $j=0$. In particular, the

integration by parts in the classical proof goes as follows:

$$\begin{aligned}
\int_0^s f(z)e^{s-z}dz &= f(s) - f(0)e^s + \int_0^s f'(z)e^zdz \\
&= (f(s) + f'(s)) - (f(0) + f'(0))e^s + \int_0^s f''(z)e^zdz \\
&= (f(s) + f'(s) + f''(s) + \dots) - (f(0) + f'(0) + f''(0) + \dots)e^s.
\end{aligned}$$

In particular, we took $f(z) = (P(z))^N$, which vanished to order N at 0 and s , and analyzed the above quantity, accessing the higher derivatives of f at 0 and s . And the fact that $(e^z)' = e^z$ is what makes this work – every time we integrate by parts, we know that our integrand is going to change in a controlled fashion.

So at least the discretization is consistent-ish between the classical proof and Schneider-Lang – you use the fact that it satisfies a differential equation to show that taking derivative doesn't increase the height too much. But, comparing when $d = 1$ (which I think makes the most sense), the classical proof saves a factor of $N!$ that the proof of Schneider-Lang doesn't. So allowing ourselves to use j up to N^ε must be pretty powerful, and the complex analysis input is somehow leveraging the disappearing pretty well.

If anybody knows a way to make the above fact tangible (e.g. by explicitly constructing such a function in any case) I would love to hear it.

§3 Schneider-Lang

We will now prove Schneider-Lang.

Theorem 3 (Schneider-Lang 1966). *Let K be a degree d extension of $\mathbb{Q}$, and let $f_1, f_2, \dots, f_t$ be functions such that $K[f_1, f_2, \dots, f_t]$ is closed under differentiation. Suppose f_1, f_2 are algebraically independent with $f_i(z) \leq C_i e^{|z|^{\rho_i}}$. Then, there are at most $(\rho_1 + \rho_2)d$ values of z such that $f_i(z) \in K$ for all i .*

Proof. Suppose the f_i are simultaneously in K at $\omega_1, \dots, \omega_m$. We will show that $m \leq (\rho_1 + \rho_2)d$ by constructing a function

$$X(z) = \sum_{i,j} c_{i,j} f_i^j(z) g^j(z) \quad (1)$$

such that X vanishes to order N at each ω_i , and the coefficients are negligible (below $(N!)^\varepsilon$). Observe that

1. some higher order derivative (pretend this is $X^{(N+1)}(\omega_1)$ for simplicity) is nonvanishing (from algebraic independence!)
2. We have $|X^{(N+1)}(\omega_1)| \ll (N!)^{1-m/(\rho_1+\rho_2)}$; this is complex analysis exploiting the high vanishing at m points.
3. We have $|X^{(N+1)}(\omega_1)| \gg \frac{1}{(N!)^{d-1}}$; this is discretization from being algebraic with small conjugates.

When we write $A \ll B$, we can be ignoring up to a factor of C^N .

Suppose we sum over $i \leq 2\varepsilon^{-1/2} m^{1/2} N^{\rho_2/(\rho_1+\rho_2)}$ and $j \leq 2\varepsilon^{-1/2} m^{1/2} N^{\rho_1/(\rho_1+\rho_2)}$ so a $4\varepsilon^{-1}$ factor over the number of needed free variables chosen to minimize the growth rate of $X(z)$. Siegel's lemma then tells us that we may pick coefficients such that

$$B = \max c_{i,j} \ll (N!)^\varepsilon \quad (2)$$

Imagining the algebraic conjugates of $X^{N+1}(\omega_1)$, one sees that they're $\ll BN! \cdot C^N$, so Vieta's tells us that

$$|X^{N+1}(\omega_1)| \gg \frac{1}{(BN! \cdot C^N)^{d-1}}. \quad (3)$$

On the other hand, one may observe that by the maximum modulus principle,

$$\left| \frac{X^{N+1}(\omega_1)}{N!} \right| \leq \sup_{|z|=R} \frac{X(z)C^N}{\prod (z - \omega_i)^N} \ll B \cdot \frac{e^{R^{\rho_1} N^{\rho_2}/(\rho_1+\rho_2) + R^{\rho_2} N^{\rho_1}/(\rho_1+\rho_2)}}{R^{mN}}. \quad (4)$$

Taking the visually appealing $R = N^{1/(\rho_1+\rho_2)}$ gives

$$|X^{N+1}(\omega_1)| \ll \frac{N!}{(N!)^{m/(\rho_1+\rho_2)}}. \quad (5)$$

Comparing [Eq. \(3\)](#) and [Eq. \(5\)](#), we are done. $\square$